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Leak Trace: tracing partial, retyped exam-paper leaks to the college that printed them

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Version 1.0, 30 October 2026 · Live demo, code and data: see Appendix B

Abstract. When an exam paper leaks from one of the hundreds of colleges that print it, the leak usually reaches students as a handful of retyped or photographed questions, and the QR codes, serial numbers and watermarks that exam bodies print do not survive. Leak Trace gives every college a content-identical paper whose arrangement encodes a college-specific codeword in three layers: the order of interchangeable questions, labels and options; faculty-approved wordings; and, new in this paper, *load-bearing marks*, the values in numerical questions and the letter under which each right option sits, which a leaker cannot change without making the leaked answers wrong. A likelihood-ratio decoder reads a leak in any form (typed text, a photographed page read by OCR, or a bare answer key) and names a college only when the evidence makes it at least $T = 10{,}000$ times likelier than every other college and T times the number of colleges likelier than a copy from outside the system; otherwise it returns a ranked shortlist. We prove that, under our model of copies made outside the system, the chance that such a copy clears the single-college naming bar is at most $1/T$ (and $2/T$ with the separate two-college test), and that the bound holds at every moment while the lines of a leak are read one by one, so investigators may re-run the decoder as more screenshots surface. In simulations with 500 colleges, seven numbered questions name the source 92% of the time; across 135,000 honest and 148,000 adversarial leaks the decoder named an innocent college once, for a copy made outside the system; a leaked answer key with 14 correct answers is named every time; with 16 approved value sets per question, six reworded and shuffled numerical questions are named in 83% of leaks; and no innocent college was named in 171,000 such leaks. A custody design (2-of-3 secret sharing, a codebook derived from the key and a signed Merkle root sent to every principal) means no single official can frame a college or swap a paper undetected. All evaluation so far is synthetic; we state the limits and the pilot that comes next.

1 Introduction

Exam papers in India leak often, and when they do, the source is found late or never. Media reports counted 89 suspected paper leaks in the ten years to 2024, 68 of them in state exams; at least 48 papers had to be re-tested, in exams taken by at least 6.5 crore candidates [1]. NEET-UG 2026 was cancelled nine days after it was held [2]. Universities that affiliate hundreds of colleges have moved from sealed packets to password-protected PDFs that each college prints shortly before the exam: Kannur University does this for its 120 colleges [3], Bangalore University moved its B.Ed. papers to principals' logins with a one-time password 30 minutes before the exam [4], VTU sends papers online 30 minutes ahead [5], and JNTU Hyderabad lists "Electronic Distribution of Examination Papers" among its exam-branch practices [6]. That printing step is where several recent leaks began. On 2 April 2025 a principal-in-charge in Kasaragod received the Kannur PIN at 9:00 and posted four to five of the actual questions to a students' WhatsApp group at 9:04, an hour before the exam [8].

Leaks rarely show a whole page. Karnataka State Law University's Contract Law-I paper was sold in January 2025 as "two, five, or 10 questions" [9]. The QR codes, serial numbers and hidden codes that boards now print [10–13] stay behind on the paper. Sources are therefore found by luck or by police work: Kannur's leak surfaced because an exam squad caught a student copying [8]; at Telangana's agricultural university the Vice-Chancellor noticed unusually high marks and CCTV showed a student finishing in ten minutes [14]; the law-university leak needed phone-to-phone forensics [9]. Even the NEET-UG 2026 leak, spotted quickly because 45 of 108 chemistry questions matched a circulating "guess paper" word for word and 90 of 204 biology questions also matched [15], took the CBI until August to attribute, with the help of handwriting examination, to three NTA subject experts [16].

While the source is unknown, every college is a suspect and every student who wrote the paper risks a re-exam. Once Kannur's source was known, its sub-committee recommended a re-exam for that one college; the Vice-Chancellor said "it is not right for everyone to suffer for the mistake of one principal" [8]. A fast, specific and calibrated lead would let a university re-examine one college instead of a hundred,

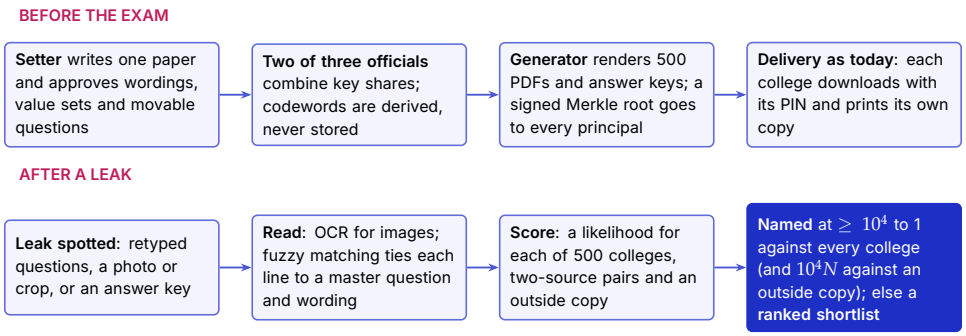


Figure 1. How Leak Trace fits into exam delivery. Only the step that makes each college's PDF changes; tracing runs only when a leak appears.

clear honest colleges quickly, and give investigators a place to start within minutes. It would also deter: an insider who knows a leak points back to their own college has a strong reason not to leak.

Contributions. This paper describes Leak Trace (Fig. 1) and its evaluation.

(1) A threat model for leaks as they actually spread: partial, retyped with typos, photographed, renumbered, reworded (including by AI tools), shuffled, mixed from two colleges, or made outside the system (§3).

(2) Content-identical fingerprints in three layers (§4). The third layer, numeric values and answer-key positions, survives rewording and shuffling because changing it makes the leaked answers wrong, a *semantic* version of the marking assumption of fingerprinting theory [17].

(3) Codes with explicit distance guarantees for each layer: a permutation code in which any two colleges share at most 4 of 10 question positions, a Reed–Solomon code over GF(23) in which they share at most one MCQ option order [18], and, as the alternative for MCQ papers whose answer keys may leak, a quaternary linear $[20, 5, 12]$ code relabelled so that every college's key is balanced.

(4) A likelihood-ratio decoder with abstention, a mixture over ambiguous numbering, a gate that counts a wording only when it is clearly matched, and a two-source stage for mixed copies (§5).

(5) A guarantee: under a stated model of copies made outside the system, the probability that such a copy clears the naming bar is at most $1/T$ per naming test, at every point of a sequential read (§6), checked on 800,000 simulated outside copies.

(6) An evaluation on 500 simulated colleges: whole-paper simulations, an end-to-end OCR test on synthetic phone photos, a 148,000-leak adversarial sweep, mixed-copy tests and 171,000 leaks of the third layer (§8).

(7) A custody and deployment design that fits today's per-college PDF delivery and ensures no single official can frame a college or swap a paper undetected (§7, §9).

Scope. Leak Trace attributes leaks that happen *after* per-college versions are generated. A leak from the setters or the press, as in NEET-UG 2026 [16], cannot be pinned on a college. For answer keys the master key, which no college receives, is itself a candidate, so such a leak points investigators upstream (§4.3); for question text the decoder currently abstains.

2 Background and related work

Canary traps and fictitious entries. Giving each recipient a slightly different copy and reading the copy that leaks is an old idea, popularised as the canary trap [19]. Content-level traps are common: the lyrics site Genius alternated straight and curly apostrophes to spell "red handed" in Morse code and found its text reproduced more than 100 times [20]; map makers insert fictitious towns such as Agloe, New York [21]; and dictionaries insert invented words [22]. These marks prove copying of an exact text. They do not survive retyping, rewording or partial copying, and they come with no error rate.

Document and text watermarking. Shifting lines and words by a fraction of a point survives photocopying [23], and modern systems match per-copy visual changes from photos [24]; retyping removes both. Natural-language watermarks change wording or syntax [25–27], and watermarks for language-model output bias the choice of words [28, 30]. All of these are fragile to paraphrase: a paraphraser dropped the detection accuracy of DetectGPT from 70.3% to 4.6% at a 1% false-positive rate [31], and after strong human paraphrasing a watermark needed about 800 tokens to be detected at a 10^{-5} false-positive rate [29]. An exam question is a sentence or two. This is why Leak Trace does not rely on wording alone.

Fingerprinting and traitor tracing. Traitor tracing assigns users keys or codewords so that leaked material identifies at least one culprit [32, 33]. Boneh and Shaw formalised collusion-secure fingerprinting under the *marking assumption*, that colluders can change a mark only where their copies differ [17]. Tardos codes achieve length $O(c^2 \log(n/\varepsilon))$ for c colluders, optimal up to a constant [34], and symmetric versions shorten them to about 5% of the original length for large coalitions [35]. Minimum-distance (traceability) codes give deterministic guarantees: if $d > n(1 - 1/c^2)$ the codeword closest to any mixture of c codewords belongs to a colluder [36]. Our setting differs from the classical one in three ways: codes are very short (a paper has 10–40 markable positions), most positions are erased (a leak shows a few questions), and reads are noisy (typos, OCR, rewording). We therefore use short structured codes, a calibrated likelihood decoder and abstention rather than Tardos-length codes.

Exam security in practice. Exam bodies in India print QR codes and serial numbers on every page [10, 12], hidden codes readable from a photo [11] or centre-code watermarks [13, 37], issue a few reordered booklets (four codes in NEET), and log who downloaded a file [38]. Vendors propose watermarks that "trace the exact source centre" [39]. In the UK, Pearson reported narrowing a 2019 leak to a single school with undisclosed tracking technology [40]. Each of these marks sits on the page, not in the content, and is lost when the questions are retyped; a few booklet codes cannot single out one college among hundreds. Variant questions as a leak-tracing "version identifying code" appear in an expired IBM patent [41], and substituting synonyms or numbers per test-taker in an abandoned application [42]; neither gives a decoder for partial, retyped leaks or an error bound. The closest academic proposals hide a logo in visual-cryptography e-papers [43] or match a leaked paper exactly against blockchain-signed sets to identify a setter [44]. We found no prior system that traces retyped partial leaks or answer keys with a stated false-accusation rate, and we claim novelty only for that combination.

Psychometrics of variant forms. Reordering is not automatically neutral. Block order had "little or no effect" on 2,940 bar candidates [45], and rearranging MCQ options showed "very small differences" at ETS [46]; but scrambled questionnaires score lower than logically ordered ones [47], an item scores higher after a related item [48], and easiest-first order gives the most correct answers [49]. The position of the right option matters: test-takers favour middle positions by up to 3 or 4 to 1 [50], and in one large study students were 1.18 times likelier to pick the right answer when it sat at A rather than E [51]. Numerical variants of an item can differ in difficulty [52]: in one study 20 of 50 item models had a facility difference above 0.15 [53]. Automatic item generation [54] supplies the method for writing and calibrating such variants. Section 9 turns these findings into constraints.

Sequential evidence. Wald's sequential probability ratio test reads observations one at a time [55]. Ville's inequality bounds the chance that a nonnegative martingale ever exceeds a level, and "the likelihood ratio is the natural test martingale" [56, 57]. We use this to make the naming bar valid however often investigators re-run the decoder.

3 Setting and threat model

Actors. A university's controller of examinations generates one paper per exam centre; we write *college* for the centre that prints a copy. JNTUH seats students at host colleges [7], so the code belongs to the printing centre, not to a student's own college. A setter writes the paper and approves its variants. A leaker inside one college sees that college's copy. Investigators receive whatever surfaces.

What leaks. (i) Text: a few questions retyped into a chat, with typos, sometimes numbered as printed, renumbered 1, 2, 3 or unnumbered. (ii) Images: a photo, crop or screenshot of the printed page. (iii)

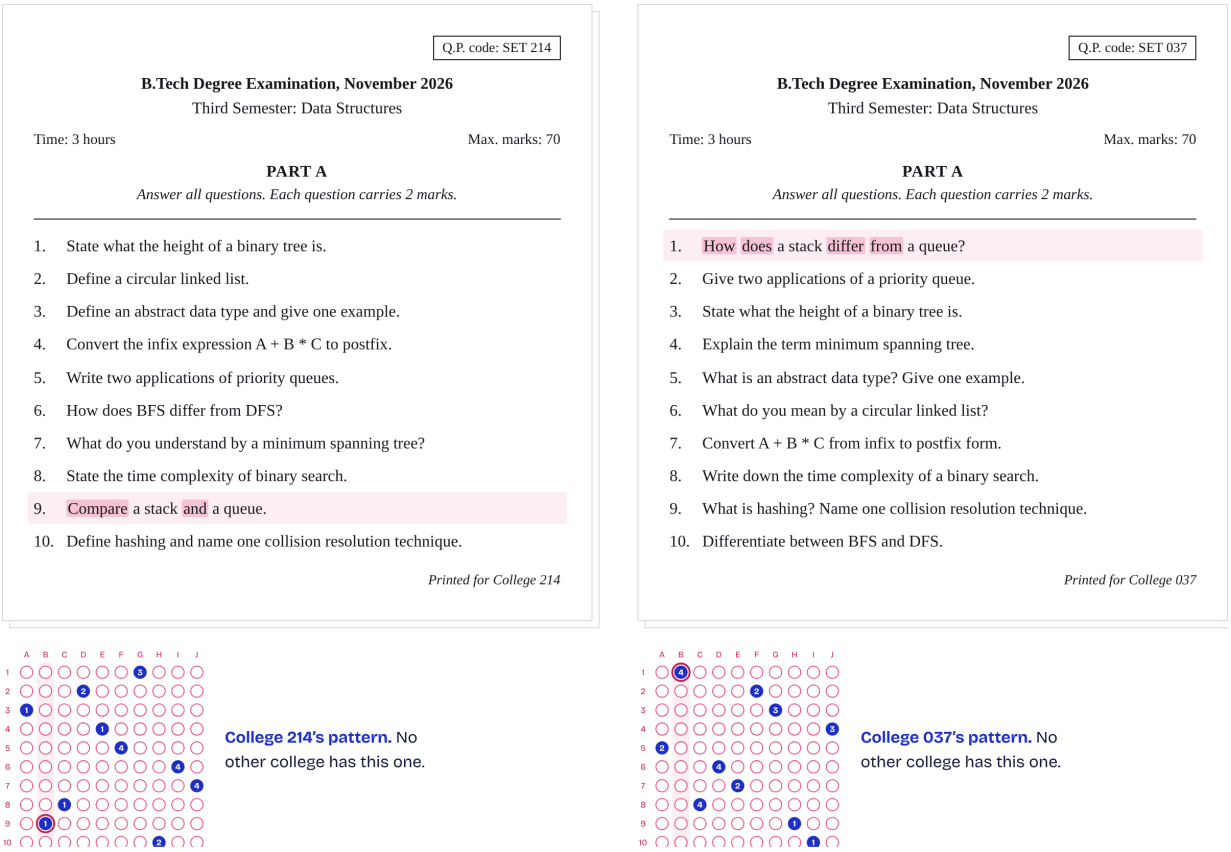


Figure 2. Part A of the same Data Structures paper as printed for two colleges. The same ten questions, marks and difficulty; a different order and a different approved wording for some questions (highlighted for question B). The grids read like an OMR sheet: columns are the ten questions, rows are printed positions, and the digit in a filled bubble is the wording that college received. Screenshot of the live demo.

Answers: a key such as "1-B 2-D 3-A", or numerical answers. Answers do leak: answers were relayed by Bluetooth in the Karnataka PSI and TSPSC cases [58, 59].

Leaker capabilities. We test, in increasing strength: copying as is; retyping with typos (up to 40% of characters damaged); dropping or changing question numbers; shuffling lines; rewording every question; rewording and shuffling; mixing items from two colleges' copies; and sending a copy not derived from any college (an outside copy, such as a guess paper or a made-up key). A leak made before versions exist (at the setter or the press) is out of scope except for being recognised as upstream.

Goals. G1: name the source when the evidence is strong. G2: name an innocent college only with a small, stated probability. G3: otherwise return a short ranked list. G4: no single official can use the system to frame a college.

Assumptions. The codebook is secret: a leaker knows their own copy but not others'. Knowing the scheme does not hide a college's own code, except by laundering the content, which we evaluate. The decoder's noise model approximates reading errors; Section 6 states what the guarantee needs.

4 Content-identical fingerprints

Every college receives the same questions, marks, sections and difficulty. Within limits the setter approves, three layers vary (Fig. 2, Table 1).

Table 1. What survives each attack. Layer 3 can only be erased by changing the numbers or the answers, which makes the leak wrong.

What the leaker does	Layer 1: order	Layer 2: wording	Layer 3: numbers, answers
Retypes with mistakes	survives	survives	survives
Sends a photo or a crop	survives	survives	survives
Drops or changes question numbers	relative order survives	survives	survives
Shuffles the lines	erased	survives	survives
Rewords every question (any tool)	survives	erased	survives
Rewords and shuffles	erased	erased	survives

4.1 Layer 1: order

Equal-mark short questions are permuted per college. In our Part A of $Q = 10$ two-mark questions, college codewords form a permutation code in which any two colleges agree on at most 4 of the 10 positions; we build it greedily from random permutations with that rejection rule. Either/or labels and the order of sub-parts add binary symbols. Each MCQ shows its four options in one of $4! = 24$ orders; we use 23 of them as symbols of GF(23) and give college c the polynomial $f_c(x) = m_0 + m_1 x$, so question i shows option order $f_c(i)$. This is a Reed–Solomon code [18] with $23^2 = 529$ codewords of length up to 23 (we use 20), and any two colleges share at most one question's option order. Order marks are constrained (§9): related questions stay together and the easy-to-hard progression is kept.

4.2 Layer 2: wording

Each short question has V faculty-approved wordings that differ in form but not meaning ("Compare a stack and a queue" / "How does a stack differ from a queue?"). We used $V = 4$. Wordings must stay far enough apart that a typo cannot turn one into another; the decoder counts a wording only when a line is clearly closer to one wording than to the others (§5).

4.3 Layer 3: load-bearing marks

Layers 1 and 2 can be laundered by a leaker who rewords and shuffles every question. Layer 3 sits in content the leak cannot do without.

Answer keys. The letter under which each right option sits varies by college. College answer keys are codewords of a quaternary linear $[20, 5, d]$ code, which we found by local search over systematic generator matrices: any two of its 1,023 nonzero codewords differ in at least $d = 12$ of 20 positions. Because a bijection of the four letters applied at one position preserves every pairwise distance, we search per-position letter labellings for which as many keys as possible are balanced, and give the 500 colleges keys in which every letter appears 3 to 7 times out of 20 and 8 to 12 right answers sit in the middle positions B and C, the positions test-takers favour [50]. The zero codeword becomes the setter's master key, which no college receives; every college key differs from it in at least 12 answers, so a leaked key that matches the master points upstream. A random codebook of the same size has closest pairs only 6 answers apart. Because an option order fixes the letter of the right answer, a paper's MCQs use either the Reed–Solomon option-order code (best when options leak with the questions) or the answer-key code (best when keys leak); we evaluate them separately. A product code, the key letter times the six orders of the three distractors, would carry both and is future work, as is deriving the key and numeric codes from the split key in our custody module.

Numeric values. For a numerical question the setter approves A sets of values chosen to be of equal difficulty (for example two quantities with four approved values each, $A = 16$), and answer keys are computed for each set. College c receives one set per question from a code with a minimum-distance filter: with 8 numerical questions, any two colleges share at most 2 sets when $A = 16$ and at most 1 when $A = 64$. A reworded leak must keep its numbers, or its answer changes; numeric answers leaked on their own carry the same symbol.

A semantic marking assumption. Fingerprinting theory assumes colluders cannot alter marks where their copies agree [17]. Layer 3 replaces that with a property of the content: a leak that changes the values

Table 2. Evidence per leaked item against an outside copy (nats, when the item matches), the smallest leak that could clear 15.4 nats, and the share of simulated leaks of that size that were named.

Leaked item	Evidence per item	Items needed (bound)	Named at that size
Numbered Part A line (position + wording)	$\ln(0.98{\cdot}10) + \ln(0.95{\cdot}4) = 3.62$	5	99.9% (demo decoder)
MCQ with its options (option order)	$\ln(0.98{\cdot}24) = 3.16$	5	90%
Answer letter, decoder allowing 1 error in 5	$\ln(0.8{\cdot}4) = 1.16$	14	100% (no errors)
Numeric value set, $A = 16$	$\ln(0.95{\cdot}16) = 2.72$	6	83%
Numeric value set, $A = 64$	$\ln(0.95{\cdot}64) = 4.11$	4	89%

or the letters of the right answers gives students wrong answers, which defeats the purpose of most leaks. Layer 3 does not stop a leaker from adding noise, which the decoder tolerates, and it does not survive a leak rebuilt from memory without numbers; it makes laundering expensive.

How much evidence a leak carries. The decoder names a college only with evidence of at least $\ln(TN) = \ln(10^4 \cdot 500) \approx 15.4$ nats (22.3 bits) against an outside copy. Table 2 divides that by the evidence one item carries under the noise model and compares the result with the smallest leak named in simulation. The agreement suggests the decoder uses most of the evidence a leak contains.

5 The decoder

Reading. Photos go through OCR (Tesseract [60] in the prototype). Each line is parsed for a leading number ("Q1.", "1)", "1 -"). A line is matched to a master question in two steps: character-bigram Dice similarity picks the two most likely questions, then the longest-common-subsequence ratio against each of their approved wordings decides, with a minimum of 0.6. A wording counts as evidence only if the best ratio is at least 0.8 and beats the nearest other wording of the same question by $\min(0.05,\ 0.4\,(1-s))$, where s is the similarity between those two approved wordings; otherwise the line counts as its question but not as a wording. Answer keys are parsed as number–letter pairs.

Likelihoods. For college c and a numbered line of question q printed at position p with wording v,

$$P_c(p, v \mid q) = \Big[(1-\varepsilon)\,\mathbb{1}\{p = \pi_c(q)\} + \tfrac{\varepsilon}{Q-1}\,\mathbb{1}\{p \neq \pi_c(q)\}\Big]\Big[(1-\varepsilon_V)\,\mathbb{1}\{v = w_c(q)\} + \tfrac{\varepsilon_V}{V-1}\,\mathbb{1}\{v \neq w_c(q)\}\Big], \quad (1)$$

with $\varepsilon = 0.02$ and $\varepsilon_V = 0.05$. For unnumbered lines the order of each pair of lines counts: $1-\varepsilon_s$ if college c prints them in that order, $\varepsilon_s = 0.03$ otherwise. Answer letters use $1-\delta$ for a match and $\delta/3$ otherwise with $\delta = 0.2$; numeric sets use $1-\varepsilon_n$ and $\varepsilon_n/(A-1)$ with $\varepsilon_n = 0.05$. The *outside-copy* model P_0 draws positions uniformly $(1/Q)$, wordings uniformly $(1/V)$, line orders uniformly $(1/k!)$, letters uniformly $(1/4)$ and value sets uniformly $(1/A)$. The log-likelihood of a leak, ℓ_c, is the sum over its items. If the numbers in a leak simply run $1, 2, \ldots, m$ with $m < Q$, they may be the leaker's own count, so both P_c and P_0 become mixtures: with probability $1-\rho$ the numbers are printed positions, with probability $\rho = 0.1$ only the order of the lines counts.

Decision. Let $\Lambda_c = e^{\ell_c - \ell_0}$ be college c's likelihood ratio against an outside copy and $\tau = \ln T$ with $T = 10^4$.

Stage 1, one source. Name $\hat{c} = \arg\max_c \ell_c$ if $\ell_{\hat{c}} - \ell_{c'} \geq \tau$ for every other college c', $\ell_{\hat{c}} - \ell_0 \geq \tau + \ln N$, and no pair of colleges explains the leak e^{τ} times better than $\hat{c}$ alone.

Stage 2, two sources. Score every pair $\{a, b\}$ (including $a = b$) with a mixture in which each item comes from a or b with probability $\frac{1}{2}$. Name a member m of the best pair if that pair beats the outside-copy model by $\tau + \ln \frac{N(N+1)}{2}$, beats every pair without m by τ, and m's partner's share could not come from an outside copy (a margin of $\tau + \ln N$).

Otherwise abstain and rank colleges by posterior probability under a uniform prior.

Stage 1 costs $O(Nk)$ and stage 2 $O(N^2k)$ for a leak of k items; for $N = 500$ that is 125,250 pairs, under a second in browser JavaScript. Answer keys use stage 1 with the master key as an extra candidate, $N+1$ in all; numeric value sets use stage 1.

6 Guarantees

Proposition 1 (Outside copies). *If a leak X is drawn from the outside-copy model P_0, then for any $T > 0$*

$$\mathbb{P}_0\Big(\exists c \in \{1, \dots, N\} : \ \Lambda_c(X) \geq NT\Big) \leq \frac{1}{T}.$$

Proof. For each c, $\mathbb{E}_0[\Lambda_c(X)] = \sum_x P_0(x)\, P_c(x)/P_0(x) \leq \sum_x P_c(x) \leq 1$ (the order model is sub-normalised). By Markov's inequality $\mathbb{P}_0(\Lambda_c \geq NT) \leq 1/(NT)$, and a union bound over the N colleges gives $1/T$. □

Proposition 2 (Anytime validity). *Let the items of a leak be read one at a time, $X_1, X_2, \dots$, and let $P_c(\cdot \mid X_{1:t-1})$ and $P_0(\cdot \mid X_{1:t-1})$ be the decoder's sequential models. If the leak follows P_0, then*

$$\mathbb{P}_0\Big(\exists t\ \exists c : \ \Lambda_c(X_{1:t}) \geq NT\Big) \leq \frac{1}{T}.$$

Proof. Each $\Lambda_c(X_{1:t})$ is a nonnegative P_0-martingale with initial value 1 [57], so their average

$$M_t = \frac{1}{N} \sum_{c=1}^{N} \Lambda_c(X_{1:t})$$

is too. Ville's inequality [56] gives $\mathbb{P}_0(\sup_t M_t \geq T) \leq 1/T$, and if some $\Lambda_c(X_{1:t}) \geq NT$ then $M_t \geq T$. □

Investigators may therefore re-run the decoder each time another line or screenshot turns up without raising the chance of naming a college for an outside copy above $1/T$. Mixtures (over numbering readings, or over error rates) are proper likelihoods, so both propositions apply to them. Stage 2's pair test uses $N(N+1)/2$ pairs in place of N and is bounded by $1/T$ in the same way, so for the full two-stage decoder a union bound gives $2/T$; answer keys use stage 1 only.

Proposition 3 (Separation by code distance). *Suppose a leak reproduces college a's marks without error on k items. (i) MCQs with options, Reed–Solomon code: every other college matches at most one item, so $\ell_a - \ell_b \geq (k-1) \ln \frac{23(1-\varepsilon)}{\varepsilon} = 7.03\,(k-1)$ nats, which exceeds τ for $k \geq 3$. (ii) Answer keys from the $[20, 5, 12]$ code with $n = 20$ answers: any other key disagrees with a's on at least $k - 8$ of the k leaked answers, so $\ell_a - \ell_b \geq (k-8) \ln \frac{3(1-\delta)}{\delta} = 2.48\,(k-8)$ nats, which exceeds τ for $k \geq 12$; each wrong answer in the leak costs at most $2 \ln 12 = 4.97$ nats of that margin.*

Proof. (i) Two Reed–Solomon codewords $m_0 + m_1 x$ agree on at most one x. (ii) Two keys differ in at least 12 of 20 positions, of which at most $20 - k$ lie outside the leak. Each disagreement contributes $\ln \frac{1-\delta}{\delta/3}$ to $\ell_a - \ell_b$ and each agreement 0; an error turns one term from $+\ln 12$ to at least $-\ln 12$. □

What is and is not guaranteed. Propositions 1–2 bound wrong names for outside copies under the outside-copy model; Section 8.7 checks them empirically for numbered Part A lines and answer keys only, and the attack test of Section 8 covers the full decoder on outside copies (1 of 32,000 named). They do not bound misattribution between colleges. For a leak that did come from some college, the runner-up test makes the named college at least T times likelier than every other; under a uniform prior over colleges its posterior probability is then at least $1/(1 + (N-1)/T) \geq 95.2\%$ for $N = 500$, and in practice far higher because most colleges are excluded by many orders of magnitude. A frequentist bound per wrong college, $\mathbb{P}_a(\Lambda_b/\Lambda_a \geq T) \leq 1/T$, sums only to $(N-1)/T \approx 5\%$, which is why the codes' distances matter (Proposition 3) and why we report misattribution empirically. Requiring TN against the runner-up as well would make that bound $1/T$ at the cost of about one or two more leaked lines. The guarantee also assumes the outside-copy model is right; Section 8.7 shows one place where real copies differ from it.

7 Custody and integrity

A tracing system is only as trustworthy as the people who hold its keys. Our design removes single points of misuse (implemented in `custody.py`, 12 of 12 self-checks pass).

(a) **No single key-holder.** The master key is split into three shares with Shamir's scheme over the prime field $2^{521} - 1$ [61]; any two officials (for example the Controller of Examinations, the Registrar and an outside observer) can rebuild it, and one alone learns nothing.

(b) **No stored table.** Every college's codeword is derived from the key with HMAC-SHA256 [62] as the random generator while keeping each code's structure (at most 4 shared positions; at most one shared option order). The college-to-pattern table exists only while two share-holders are present; deriving all 500 codewords takes about 0.6 s.

(c) **A fingerprint that leaves the building.** All per-college PDFs are hashed into one Merkle root [63] with RFC 6962 leaf and node prefixes [64], signed by two officials with Ed25519 [65] and sent to every principal before the exam. Each college can later prove which file it received, and nobody can swap a paper afterwards without the published root exposing it. No single official can therefore frame a college or swap a paper undetected; two colluding key-holders could still leak a chosen college's paper, which is why the log and the inquiry, not the decoder alone, decide.

(d) **A hash-chained log** of every generation and decoding, for audit.

An attribution is a lead, not a verdict. It should be reported with its likelihood ratios and the features that matched, used to decide which phones and logs to examine, and confirmed by other evidence; used in court it would be expert evidence about an electronic record under the Bharatiya Sakshya Adhiniyam, 2023.

8 Evaluation

8.1 Setup

All experiments use $N = 500$ simulated colleges, $T = 10^4$ ($\tau = 9.21$ nats) and fixed seeds. The paper is a Data Structures paper: Part A has ten two-mark questions with four approved wordings each; Part B has either/or pairs with two sub-parts; 20 MCQs carry option orders. Typing damage deletes, replaces or doubles characters at 4% unless stated, and half the lines are lower-cased with punctuation removed. Code and outputs are in the public repository (Appendix B).

8.2 Whole-paper simulation

Figure 3 shows the share of leaks named, and the share with the real source in a top-5 shortlist, by number of leaked items (3,000 leaks per point). Numbered leaks with wording variants are named from 5 questions in 36% of leaks, 6 in 74%, 7 in 92% and 8 in 97%; from 4 questions the source is in the top 5 in 99.9%. MCQs with options are named from 5 questions in 90%. Unnumbered leaks need more: 51% at 8 questions and 84% at 10 with wording variants. These leaks draw items from the whole paper, Part A and Part B; Part B items carry less evidence each, so they need more items than the Part-A-only leaks of Table 3. No honest single-source leak was pinned on an innocent college in 135,000 leaks, and none of 135,000 outside copies was named.

8.3 End to end, through OCR

We rendered college pages, damaged them like phone photos forwarded on WhatsApp (crops showing 3–6 questions, rotation, perspective, blur, noise, downscaling and JPEG compression) and decoded them through Tesseract. Of 80 photos, 99.4% of visible questions were recovered. All 30 photos showing 5 or 6 questions were named, all correctly; photos showing 4 questions were never named but had the source in the top 5 every time (26 of 26), and 3-question photos in 96% (23 of 24). No photo was pinned on an innocent college.

8.4 Adversarial sweep

We attacked the decoder that runs in the public demo with 148,000 simulated leaks in 74 settings (Table 3). Tricks can stop it naming anyone; they did not make it name an innocent college, with one exception: an outside copy of six numbered lines whose positions all matched one college and whose wordings matched

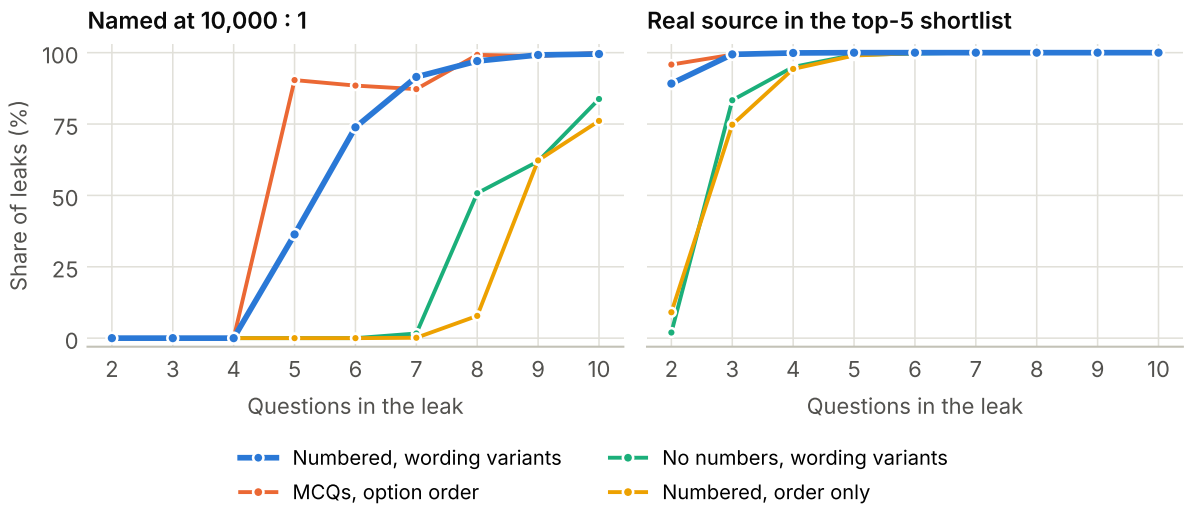


Figure 3. Whole-paper simulation, 500 colleges, 3,000 leaks per point. Left: leaks named at 10,000 to 1. Right: leaks whose real source is in the top-5 shortlist (ties counted against us).

Table 3. Adversarial sweep of the demo decoder, 2,000 leaks per setting, typing damage at 4% of characters except in the damage row. "Named from" is the smallest leak named in at least 95% of cases.

What the leaker did	Named from	In top 5, 6 lines	Innocent named
Kept the printed numbers	5 lines	100%	0 of 12,000
Left the numbers out	7 lines	100%	0 of 12,000
Renumbered the lines 1, 2, 3	7 lines	100%	0 of 12,000
Copied the first questions (real numbers 1, 2, 3)	6 lines	100%	0 of 8,000
Reworded every question, kept numbers	holds back (61% at 10)	99.8%	0 of 12,000
Reworded every question, no numbers	holds back	56.3%	0 of 12,000
Shuffled the lines, no numbers	holds back	31.4%	0 of 12,000
Mixed two colleges' copies, numbered	10 lines	>99.9%	0 of 6,000
Mixed two colleges' copies, no numbers	holds back (3% at 10)	88.8% (10 lines)	0 of 2,000
Sent a copy from outside the system	–	–	1 of 32,000
Typing damage 0–40%, 6 lines (14 settings)	see text	≥99.9%	0 of 28,000
All 74 settings			1 of 148,000

five of six. Before we hardened the decoder (weighing both readings of numbers that run 1, 2, 3; gating wordings; raising the bar from 1,000 to 10,000), the same leaks named eight innocent colleges. A separate count over one million random outside copies per length bounds the chance that some college clears the outside-copy bar at about 1 in 20,000 (6 lines), the peak over lengths. Under heavier typing damage, 5 numbered lines are still named in 99.2% of leaks at 15% character damage and 95.8% at 20%; 7 unnumbered lines in 99.7% and 93.2%. No innocent college was named in any of the 70,000 leaks of the damage tests (0–40% of characters damaged).

8.5 Mixed copies

When each leaked item comes at random from one of two colleges' copies, the two-stage decoder of the Python reference names only real sources in 59% of 10-question numbered leaks and 87% of 12-question ones, and in 83% of 10-MCQ leaks; with fewer items it mostly abstains, with a real source in the top 5 for 88–100%. No innocent college was named in 45,000 mixed leaks, 15,000 single-source leaks or 15,000 outside copies with both stages on. At the old 1,000-to-1 bar two mixes framed a third college whose random Part A wording variants happened to match; structured codes are the fix. For MCQs, the Reed–Solomon code rules framing out for two colluders once five or more MCQs leak without misreads, the traceability condition $d > n(1 - 1/c^2)$ [36], and it named no innocent college in 26,000 leaks (nor did a random codebook in another 26,000). Part A's ten positions are too few to meet that condition, so a structured code for wordings is future work.

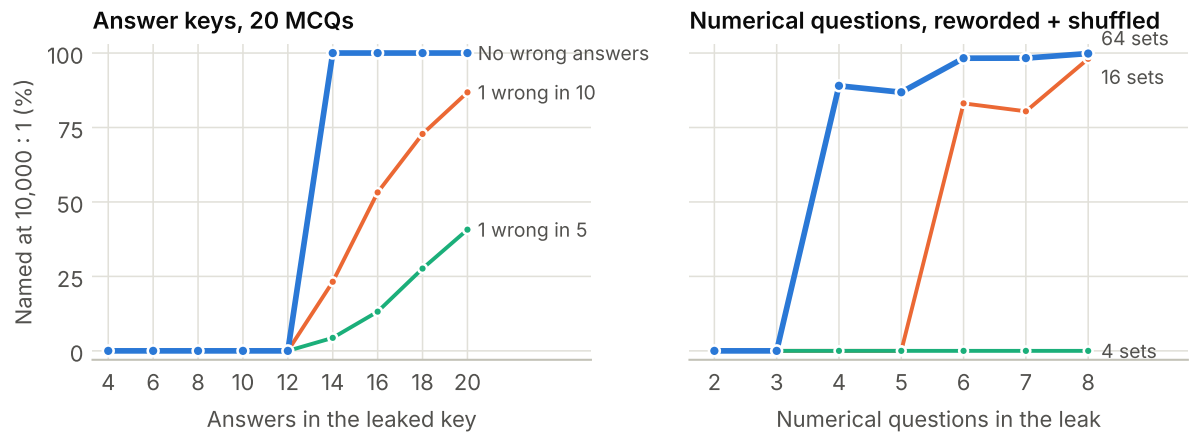


Figure 4. Layer 3, 500 colleges, 3,000 leaks per point. Left: answer keys from the $[20, 5, 12]$ code, by number of leaked answers and share of wrong answers. Right: numerical questions reworded and shuffled so that only their values survive, by number of approved value sets per question.

Table 4. Layer 3 summary (3,000 leaks per cell). Top 5 counts ties against us.

Leak	Named	Real source in top 5	Innocent named
Answer key, 14 answers, no errors	100%	100%	0
Answer key, 20 answers, 1 error in 10	86.8%	100%	0
Answer key, 8 answers, 1 error in 10	0%	95.4%	0
Answer key, 20 answers, 1 error in 5	40.7%	99.9%	0
Setter's master key, 16 answers, 1 error in 10	traced upstream 52.1%	–	0
Numeric, 16 sets, 6 questions, reworded and shuffled	83.1%	100%	0
Numeric, 16 sets, 8 questions	98.1%	100%	0
Numeric, 64 sets, 4 questions	89.0%	>99.9%	0
All 171,000 layer-3 leaks (code keys and numeric)			0

8.6 Layer 3: answer keys and numeric values

Figure 4 and Table 4 summarise 3,000 leaks per point. A key with no wrong answers is named from 14 answers every time; with one wrong answer in ten, 20 answers are named in 87% of keys; with one in five, 41%. Below those sizes the source stays near the top: from 8 answers with one error in ten it is in the top 5 for 95.4% of keys. Leaks of the master key (16 answers, one error in ten) were traced upstream in 52% and never pinned on a college. The random codebook names about as often but its closest keys are only 6 answers apart, against 12 for the code. For numeric values, after rewording and shuffling so that only the numbers survive (3% of numbers misread), 16 approved sets per question name the source from 6 questions in 83% of leaks and from 8 in 98%; 64 sets name it from 4 questions in 89%. With 4 sets nothing is named, but the source is in the top 5 for at least 98% of 6-question leaks. Naming is not monotone at the margin (83% at 6, 80% at 7) because one misread value costs about two questions' worth of evidence ($2.72 + 2.93 = 5.65$ nats), so these rates are close to 0.97^6 and 0.97^7, the chance that no value is misread. No innocent college was named in 171,000 layer-3 leaks.

8.7 The guarantee, checked

We read 800,000 outside copies one item at a time: 200,000 numbered Part A copies drawn from the decoder's own outside-copy model, 200,000 realistic Part A copies and 400,000 answer keys, and counted how often *any* college's ratio ever reached TN (Fig. 5). This checks the single-college test for these two channels; the attack test above covers the full decoder. For outside copies drawn from the decoder's own model the rate stays under $1/T$ at every bar (5×10^{-6} at $T = 10^4$). For realistic copies, whose printed positions form a permutation rather than independent draws, the rate is 5.5×10^{-5} at $T = 10^4$ (11 of 200,000), inside the bound, but at $T = 100$ it is 1.13%, slightly above the 1% the model promises, because positions drawn without replacement match a college's pattern a little more often than independent ones. Answer

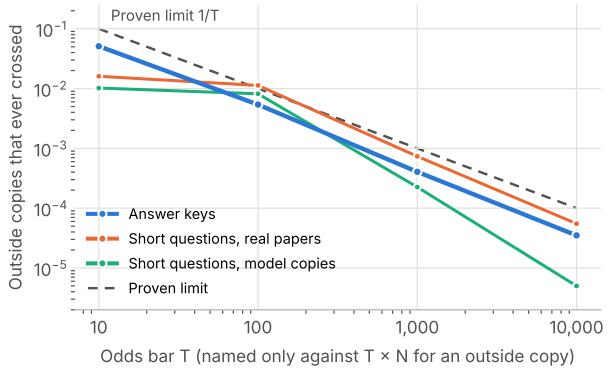


Figure 5. Share of outside copies for which some college's likelihood ratio ever reached $T \times N$ while the copy was read item by item, against the proven limit $1/T$ (Proposition 2). Model copies follow the decoder's outside-copy model; real papers draw positions as a permutation, which exceeds the model's bound slightly at $T = 100$. At the bar we use, $T = 10^4$, all three stay inside it.

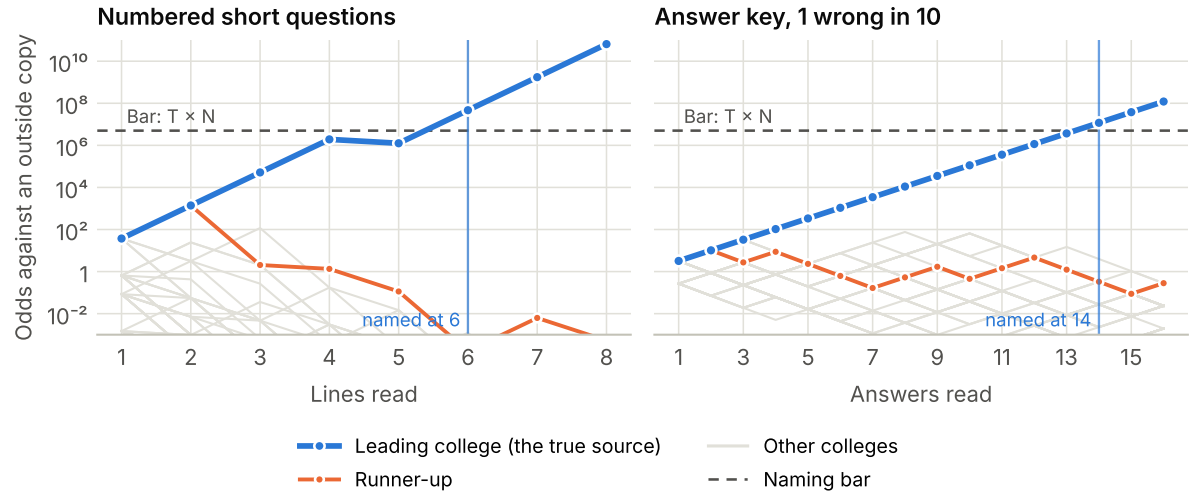


Figure 6. Evidence growing as a leak is read. Left: eight numbered short questions from one college; the source crosses the bar and leads every other college by 10^4 at line 6. Right: a 16-answer key with one wrong answer in ten; named at answer 14. Grey lines are the next 38 colleges.

keys stay inside the bound everywhere (3.5×10^{-5} at $T = 10^4$, 14 of 400,000). At $T = 10^4$ the realistic Part A rate is within a factor of two of the bound, so the model matters. Crossing the outside-copy bar is necessary but not sufficient to be named, since the runner-up test must also pass. Figure 6 shows two sequential reads.

8.8 Runtime

Decoding a leak against 500 colleges, including the $N^2/2$ pair stage, takes well under a second in browser JavaScript on a laptop; the live demo runs 200 leaks in a few seconds. Deriving 500 codewords from the key takes about 0.6 s in Python.

9 Fairness and deployment

Fairness constraints. Following the evidence in Section 2: we permute only within equal-mark blocks, either/or pairs, sub-parts and MCQ options, where we expect order to matter least; keep related questions adjacent and the easy-to-hard order fixed; have a faculty member approve every wording and value set; balance every answer key (3–7 of each letter, 8–12 middle answers); and generate answer keys per version so valuers mark against the right key, with the version ID printed for them. Numeric value sets must be equivalent in steps and arithmetic, and because facility can still differ [53], a pilot should calibrate them before high-stakes use. After every exam the university publishes mean marks per version; a version that scores differently is visible at once.

Fit with delivery. Only one step changes: the moment a university makes each college's PDF. Colleges download and print as today. For computer-based tests, which NEET moves to from 2027 [66], a different arrangement per centre or per candidate costs nothing, and a phone photo of a screen is traced the same way. Last-minute generation from a verified question bank, as recently proposed [67, 68], is compatible: per-centre versions are made at the same moment.

Side effect. Because any two colleges' answer keys differ in at least 12 of 20 answers, a key leaked from one college is mostly wrong at every other college. This contains the damage of the answer relays seen in recruitment exams [59], independently of tracing.

Cost and pilot. The software is open source and runs on a laptop; the main recurring cost is setter time to write two or three wordings or value sets per question. Our plan: a campus leak lab and student survey (November 2026), a per-college PDF generator with answer keys (December), a pilot in one college's internal exams with one version per hall (next semester), then one university exam series.

10 Limitations

- **Synthetic evaluation.** Pages, damage and leaker behaviour are simulated. How often real leakers renumber, reword or shuffle (ρ and the attack mix) is unknown until the leak lab.
- **Upstream leaks.** Leaks at the setters or the press, as in NEET-UG 2026 [16], cannot be pinned on a college. For answer keys Leak Trace recognises them as matching the master key; for question text it abstains.
- **Laundered leaks.** A leaker who rewords and shuffles every short question erases layers 1 and 2. Without numerical questions or an answer key the decoder abstains; it does not guess. A paper rebuilt from memory, like a "guess paper", keeps little structure.
- **Answer keys as text.** A key written as answer texts instead of letters loses the letter channel, though the option texts then carry layer 1.
- **Collusion.** Two-copy mixes are handled; larger coalitions would need far longer codes than one paper offers [34].
- **Misattribution bound.** The $1/T$ bound covers outside copies; between colleges we give a posterior bound and empirical counts (§6).
- **Languages and OCR.** English only. Tesseract ships Telugu, Kannada, Tamil and Malayalam models, untested here; vision-language OCR may normalise wording and blur layer 2.
- **Law and practice.** No Indian court has considered a statistical attribution of a leaked paper; the output is designed as an investigative lead.

11 Conclusion

Paper leaks in India are detected quickly and attributed slowly. Leak Trace moves attribution to the moment a leak is seen, by printing each college's identity into the content of a fair paper and reading it back with a decoder that states its own error rate and refuses to guess. The third layer of marks, numbers and answers, is the strongest: a leaker cannot remove it without making the leaked answers wrong. The next step is real data: a leak lab with our classmates, then a pilot in one college's internal exams.

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A Parameters

Parameter	Value
Colleges N, naming bar T	500; 10^4 ($\tau = 9.21$ nats; outside-copy margin $\tau + \ln N = 15.4$ nats)
Part A	$Q = 10$ positions, permutation code with at most 4 shared positions; $V = 4$ wordings
Noise	position $\varepsilon = 0.02$; wording $\varepsilon_V = 0.05$; pair order $\varepsilon_s = 0.03$; own-numbering prior $\rho = 0.1$
Matching	bigram Dice for the top 2 questions, LCS ratio ≥ 0.6; wording counts if $r \geq 0.8$ and margin $\geq \min(0.05, 0.4(1-s))$
MCQs	20 questions, Reed–Solomon over GF(23), $f_c(x) = m_0 + m_1 x$
Answer keys	quaternary $[20, 5, 12]$ linear code, per-position letter labelling, 3–7 of each letter, 8–12 in B/C; $\delta = 0.2$
Numeric values	8 questions; $A \in \{4, 16, 64\}$ with at most 5, 2, 1 shared sets; $\varepsilon_n = 0.05$
Custody	Shamir 2-of-3 over $2^{521} - 1$; HMAC-SHA256; RFC 6962 Merkle tree; Ed25519

B Reproducibility

Live demo (runs the decoder in the browser):
https://raw.githack.com/SakethSumanBathini/SakethSumanBathini/leak-trace-site/index.html
Code, seeds and result files: https://github.com/SakethSumanBathini/SakethSumanBathini/tree/leak-trace-site
`sim_decode.py` (whole paper), `e2e_ocr.py` (OCR), `sim_coalition.py` (mixes), `sim_rs_mcq.py` (Reed–Solomon), `sim_tier3.py` (answer keys, numeric values, guarantee check), `adversarial/sweep.js` (attack sweep on the demo decoder), `custody.py` (key custody, 12 self-checks). Results quoted here are in `results/` and `adversarial/runs/`.

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